WI4014TU – Numerical Analysis for PDE's

Lecture 14
Linear solvers, LU factorization, classical iterative methods, regular splittings, Jacobi, Gauss-Seidel, SOR, Gerschgorin's Lemma

Why study linear solvers?

- Numerical solution of all PDE's involves solving one or multiple linear algebraic problems: Poisson eq. (one), diffusion with Backward Euler (one per outer iteration), nonlinear equations with BE (one per inner iteration), etc
- Matrix assembly and linear solvers are computationally expensive
- Large computational domains and fine spatial discretization lead to very large linear algebraic problems
- Some system matrices are symmetric, other are not and may require special linear solvers

Well-conditioned linear problems

We are considering linear systems of the form

$$A\boldsymbol{x} = \boldsymbol{b}$$

where

- $A \in \mathbb{R}^{n \times n}$, $\boldsymbol{x}, \boldsymbol{b} \in \mathbb{R}^n$
- A is *nonsingular* and *well conditioned*
- we shall see how other properties of A help to find a suitable solver

A few words about direct solvers

LU factorization

- A well conditioned system $A\boldsymbol{x} = \boldsymbol{b}$ can, in principle, be solved by Gaussian elimination without pivoting
- A practical way of doing Gaussian elimination is to factorize the matrix A as:

$$A = LU$$

  where U is upper triangular and L is lower-triangular with all diagonal elements equal unity
- Then, the linear system $A\boldsymbol{x} = \boldsymbol{b}$ is replaced by two systems $L\boldsymbol{y} = \boldsymbol{b}$ and $U\boldsymbol{x} = \boldsymbol{y}$ both of which can be solved by simple substitution in $\mathcal{O}(n^2)$ Floating-Point OPerations (FLOP's)
- LU factorization itself costs $\mathcal{O}(n^3)$ operations:

$$u_{k,j} = a_{k,j} - \sum_{i=1}^{j-1} \ell_{k,i} u_{i,j}, \quad j = k, k+1, \ldots, n$$

$$\ell_{j,k} = \frac{1}{u_{k,k}} \left(a_{j,k} - \sum_{i=1}^{k-1} \ell_{j,i} u_{i,k} \right), \quad j = k+1, k+2, \ldots, n$$

LU factorization of banded matrices

- Matrix A with elements $a_{k,m}$ is called *banded* with *bandwidth* s if $a_{k,m} = 0$ for every $k, m \in \{1, 2, \ldots, n\}$ such that $|k - m| > s$.
- Example: tri-diagonal matrix has bandwidth $s = 1$
- Ordering of unknowns influences the bandwidth of A
- **Storage (elements):** dense – n^2, banded $\sim (2s + 1)n$
- **Operations count (flops):**
  - LU factorization: dense – $\mathcal{O}(n^3)$, banded – $\mathcal{O}(s^2 n)$
  - Solving $L\boldsymbol{y} = \boldsymbol{b}$ and $U\boldsymbol{x} = \boldsymbol{y}$: dense – $\mathcal{O}(n^2)$, banded – $\mathcal{O}(sn)$
- If the matrix A is positive definite, then a twice cheaper 'perfect' Cholesky factorization $A = LL^T$ can be applied if the ordering of unknowns is right (tree structure)
- Solver based on factorization of A is called a *direct solver*

Classical Iterative Methods

Linear one-step stationary iterative scheme

Consider the system

$$Ax = \boldsymbol{b} \tag{1}$$

and the iterative procedure:

$$\boldsymbol{x}_{k+1} = H\boldsymbol{x}_k + \boldsymbol{v} \tag{2}$$

The *spectral radius* $\rho(A) = \max\{|\lambda| : \lambda \in \sigma(A)\}$, where $\sigma(A)$ is the set of eigenvalues (*spectrum*) of A.

It is always true that $\rho(A) \leq \|A\|$, where $\|\cdot\|$ is the Euclidean norm. For *normal* matrices, i.e., such that $A^T A = AA^T$, $\rho(A) = \|A\|$. For general square matrices: $\|A\| = [\rho(A^T A)]^{1/2}$

Theorem: *Given an arbitrary linear system (1), a linear, one-step, stationary scheme (2) converges to a unique bounded limit* $\tilde{\boldsymbol{x}}$*, regardless of the choice of the starting vector* $\boldsymbol{x}_0$*, if and only if* $\rho(H) < 1$*. Moreover,* $\tilde{\boldsymbol{x}}$ *is the solution of (1) if and only if* $\boldsymbol{v} = (I - H)A^{-1}\boldsymbol{b}$

Proof (1): Let $\rho(H) < 1$ and let H be diagonalizable with the eigendecomposition $H = V\Lambda V^{-1}$. Then,

$$\lim_{k\to\infty} H^k = \lim_{k\to\infty} (V\Lambda V^{-1})^k = \lim_{k\to\infty} (V\Lambda V^{-1} V\Lambda V^{-1} \cdots V\Lambda V^{-1})$$
$$= V \left(\lim_{k\to\infty} \Lambda^k \right) V^{-1} = 0$$

since $|\lambda_i| < 1$, $\lambda_i \in \sigma(H)$. If H is not diagonalizable, the same can be proven using the Jordan decomposition $H = WJW^{-1}$.

We claim that, in general:

$$\boldsymbol{x}_k = H^k \boldsymbol{x}_0 + (I-H)^{-1}(I-H^k)\boldsymbol{v}, \quad k = 0, 1, \ldots$$

Induction: (i) It is true for $k = 0$, since $H^0 = I$. (ii) Assume it holds for some $k \geq 0$. (iii) Then, it also holds for $k+1$:

$$\begin{aligned}
\boldsymbol{x}_{k+1} &= H\boldsymbol{x}_k + \boldsymbol{v} = H\left[H^k \boldsymbol{x}_0 + (I-H)^{-1}(I-H^k)\boldsymbol{v}\right] + \boldsymbol{v} \\
&= H^{k+1}\boldsymbol{x}_0 + \left[H(I-H)^{-1}(I-H^k) + I\right]\boldsymbol{v} \\
&= H^{k+1}\boldsymbol{x}_0 + \left[(I-H)^{-1}H(I-H^k) + (I-H)^{-1}(I-H)\right]\boldsymbol{v} \\
&= H^{k+1}\boldsymbol{x}_0 + (I-H)^{-1}(H - H^{k+1} + I - H)\boldsymbol{v} \\
&= H^{k+1}\boldsymbol{x}_0 + (I-H)^{-1}(I-H^{k+1})\boldsymbol{v}
\end{aligned}$$

Proof (2): Hence, for $\rho(H) < 1$, the iterative process converges to the vector $\tilde{\boldsymbol{x}}$:

$$\tilde{\boldsymbol{x}} = \lim_{k\to\infty} \boldsymbol{x}_k = \lim_{k\to\infty} H^k \boldsymbol{x}_0 + (I - H)^{-1}(I - H^k)\boldsymbol{v} = (I - H)^{-1}\boldsymbol{v}$$

Let $\rho(H) \geq 1$.
If $1 \notin \sigma(H)$, then $I - H$ is invertible and $\tilde{\boldsymbol{x}} = (I - H)^{-1}\boldsymbol{v}$ is the only possible bounded limit of the iterative scheme. Assume there was another bounded limit $\tilde{\boldsymbol{y}}$. Then,

$$\tilde{\boldsymbol{y}} = \lim_{k\to\infty} \boldsymbol{x}_{k+1} = \lim_{k\to\infty} (H\boldsymbol{x}_k + \boldsymbol{v}) = H \lim_{k\to\infty} \boldsymbol{x}_k + \boldsymbol{v} = H\tilde{\boldsymbol{y}} + \boldsymbol{v}$$

and $\tilde{\boldsymbol{y}} = (I - H)^{-1}\boldsymbol{v} = \tilde{\boldsymbol{x}}$, i.e., the limit, if it exists, is unique.

If $1 \in \sigma(H)$ and $\boldsymbol{w}$ is the corresponding eigenvector $H\boldsymbol{w} = \boldsymbol{w}$, then, whenever there is a limit, $\tilde{\boldsymbol{y}} = H\tilde{\boldsymbol{y}} + \boldsymbol{v}$, it follows that $\tilde{\boldsymbol{y}} + \boldsymbol{w} = H(\tilde{\boldsymbol{y}} + \boldsymbol{w}) + \boldsymbol{v}$ is also a limit. Hence, with $1 \in \sigma(H)$, either there is no limit or the limit is not unique, i.e., there is no convergence.

Proof (3):

Choose $\lambda \in \sigma(H)$ such that $|\lambda| = \rho(H) \geq 1$. Let now $\boldsymbol{w}$ be the eigenvector of this eigenvalue, i.e., $H\boldsymbol{w} = \lambda \boldsymbol{w}$, and let $\|\boldsymbol{w}\| = 1$. We show that in this case there is always a starting vector $\boldsymbol{x}_0$ for which the iterations will fail to converge.
Case 1: Let $\lambda \in \mathbb{R}$. Hence, $\boldsymbol{w}$ is also real (up to a multiplier). Choose the starting vector as $\boldsymbol{x}_0 = \boldsymbol{w} + \tilde{\boldsymbol{x}}$. We claim that

$$\boldsymbol{x}_k = \lambda^k \boldsymbol{w} + \tilde{\boldsymbol{x}}, \quad k = 0, 1, \ldots$$

Induction: (i) This is true for $k = 0$. (ii) Assume it is true for some $k \geq 0$. (iii) Then, it is also true for $k+1$:

$$\begin{aligned} \boldsymbol{x}_{k+1} &= H\boldsymbol{x}_k + \boldsymbol{v} = H(\lambda^k \boldsymbol{w} + \tilde{\boldsymbol{x}}) + \boldsymbol{v} = \lambda^{k+1}\boldsymbol{w} + H(I-H)^{-1}\boldsymbol{v} + \boldsymbol{v} \\ &= \lambda^{k+1}\boldsymbol{w} + (I-H)^{-1}H\boldsymbol{v} + (I-H)^{-1}(I-H)\boldsymbol{v} \\ &= \lambda^{k+1}\boldsymbol{w} + (I-H)^{-1}(H+I-H)\boldsymbol{v} = \lambda^{k+1}\boldsymbol{w} + (I-H)^{-1}\boldsymbol{v} = \lambda^{k+1}\boldsymbol{w} + \tilde{\boldsymbol{x}} \end{aligned}$$

It follows that, since $|\lambda| \geq 1$,

$$\|\boldsymbol{x}_k - \tilde{\boldsymbol{x}}\| = \|\lambda^k \boldsymbol{w}\| = |\lambda|^k \geq 1, \quad k = 0, 1, \ldots$$

In other words, the sequence $\{\boldsymbol{x}_k\}$ does not converge to $\tilde{\boldsymbol{x}}$.

Proof (4): *Case 2:* Let $\lambda \in \mathbb{C}$. Then, if $(\lambda, \boldsymbol{w})$ is an eigenpair, so is $(\overline{\lambda}, \overline{\boldsymbol{w}})$. Furthermore, since $\lambda \neq \overline{\lambda}$, the two eigenvectors, $\boldsymbol{w}$ and $\overline{\boldsymbol{w}}$, are linearly independent. Otherwise, they would correspond to the same eigenvalue. Consider the function

$$g(z) = \|z\boldsymbol{w} + \overline{z}\,\overline{\boldsymbol{w}}\|, \quad z \in \mathbb{C}$$

This is a continuous function $g : \mathbb{C} \to \mathbb{R}$. Hence, it attains a minimum on every closed and bounded subset of $\mathbb{C}$, in particular, on the unit circle. Therefore,

$$\inf_{-\pi \le \theta \le \pi} \|e^{i\theta}\boldsymbol{w} + e^{-i\theta}\overline{\boldsymbol{w}}\| = \min_{-\pi \le \theta \le \pi} \|e^{i\theta}\boldsymbol{w} + e^{-i\theta}\overline{\boldsymbol{w}}\| = \nu \ge 0$$

Suppose that $\nu = 0$. Then there exists $\theta_0 \in [-\pi, \pi]$ such that

$$\|e^{i\theta}\boldsymbol{w} + e^{-i\theta}\overline{\boldsymbol{w}}\| = 0$$

and $\overline{\boldsymbol{w}} = e^{2i\theta}\boldsymbol{w}$, i.e., $\boldsymbol{w}$ and $\overline{\boldsymbol{w}}$ are linearly dependent, which is not possible. Hence, $\nu > 0$. The function $g(z)$ is homogeneous,

$$g(re^{i\theta}) = \|re^{i\theta}\boldsymbol{w} + re^{-i\theta}\overline{\boldsymbol{w}}\| = r\|e^{i\theta}\boldsymbol{w} + e^{-i\theta}\overline{\boldsymbol{w}}\| = rg(e^{i\theta}), \quad r > 0,\ |\theta| \le \pi$$

Proof (5): In particular,

$$g(z) = g\left(|z|\frac{z}{|z|}\right) = |z|\,g\left(\frac{z}{|z|}\right) \geq |z|\,\nu, \quad z \in \mathbb{C} \setminus \{0\}$$

Choose the starting vector as $\boldsymbol{x}_0 = \boldsymbol{w} + \overline{\boldsymbol{w}} + \tilde{\boldsymbol{x}}$. It is possible to show inductively that

$$\boldsymbol{x}_k = \lambda^k \boldsymbol{w} + \overline{\lambda}^k \overline{\boldsymbol{w}} + \tilde{\boldsymbol{x}}, \quad k = 0, 1, \ldots$$

Hence,

$$\|\boldsymbol{x}_k - \tilde{\boldsymbol{x}}\| = \|\lambda^k \boldsymbol{w} + \overline{\lambda}^k \overline{\boldsymbol{w}}\| = g(\lambda^k) \geq |\lambda|^k \nu \geq \nu > 0, \quad k = 0, 1, \ldots$$

and the sequence $\{\boldsymbol{x}_k\}$ is bounded away from the only possible limit $\tilde{\boldsymbol{x}}$, i.e., it cannot converge.

The claim that $\boldsymbol{v}$ must be equal $(I - H)A^{-1}\boldsymbol{b}$ is easy to check, since $\boldsymbol{x} = A^{-1}\boldsymbol{b}$ is the exact solution and $\tilde{\boldsymbol{x}} = (I - H)^{-1}\boldsymbol{v}$ is the unique limit. □

One-step iteration: conclusion

- To solve $A\boldsymbol{x} = \boldsymbol{b}$
- One could use iterations $\boldsymbol{x}_{k+1} = H\boldsymbol{x}_k + \boldsymbol{v}$
- Provided that $\rho(H) < 1$ and $\boldsymbol{v} = (I - H)A^{-1}\boldsymbol{b}$
- Not practical, used as a tool for studying other methods

Incomplete LU factorization (ILU)

Suppose that there is a splitting $A = \hat{A} - E$ such that it is easy to find the LU factorization of $\hat{A}$ (e.g. $\hat{A}$ is banded). Then,

$$\begin{aligned} A\boldsymbol{x} &= \boldsymbol{b} \\ (\hat{A} - E)\boldsymbol{x} &= \boldsymbol{b} \\ \hat{A}\boldsymbol{x} &= E\boldsymbol{x} + \boldsymbol{b} \end{aligned}$$

This suggest the following *Incomplete LU Factorization* iterative scheme:

$$\hat{A}\boldsymbol{x}_{k+1} = E\boldsymbol{x}_k + \boldsymbol{b}$$

where one needs to compute the LU factorization of $\hat{A}$ and re-use it at each iteration to find $\boldsymbol{x}_{k+1}$.

This can be put in the form $\boldsymbol{x}_{k+1} = H\boldsymbol{x}_k + \boldsymbol{v}$ if we choose $H = \hat{A}^{-1}E$ and $\boldsymbol{v} = \hat{A}^{-1}\boldsymbol{b}$

Regular splittings

ILU, where $A = \hat{A} - E$, is an example of the *regular splitting* $A = P - N$, where P is *nonsingular*. Consider the iterative scheme:

$$P\boldsymbol{x}_{k+1} = N\boldsymbol{x}_k + \boldsymbol{b} \tag{3}$$

Theorem: *Suppose that both A and $P + P^T - A$ are real, symmetric and positive definite (SPD). Then the scheme (3) converges.*

Note that in this case $H = P^{-1}N = P^{-1}(P - A) = I - P^{-1}A$ and $\boldsymbol{v} = P^{-1}\boldsymbol{b}$. Proof shows that under these assumptions all eigenvalues λ of $I - P^{-1}A$ have $|\lambda| < 1$.

Proof (1): Let $\lambda \in \mathbb{C}$ be an arbitrary eigenvalue of $H = I - P^{-1}A$ and $\boldsymbol{w}$ – the corresponding eigenvector. Then,

$$\begin{aligned}
(I - P^{-1}A)\boldsymbol{w} &= \lambda \boldsymbol{w} \\
P(I - P^{-1}A)\boldsymbol{w} &= P\lambda \boldsymbol{w} \\
(P - A)\boldsymbol{w} &= \lambda P \boldsymbol{w} \\
P\boldsymbol{w} - \lambda P \boldsymbol{w} &= A\boldsymbol{w} \\
(1-\lambda)P\boldsymbol{w} &= A\boldsymbol{w}
\end{aligned}$$

First, we conclude that $\lambda \neq 1$, since otherwise $A\boldsymbol{w} = \mathbf{0}$ for some $\boldsymbol{w} \neq \mathbf{0}$, i.e., A is singular, which contradicts the assumption that A is SPD. Further, since A is symmetric and $\lambda \neq 1$,

$$\begin{aligned}
(1-\lambda)\overline{\boldsymbol{w}}^T P \boldsymbol{w} &= \overline{\boldsymbol{w}}^T A \boldsymbol{w} \in \mathbb{R} \\
(1-\overline{\lambda})\overline{\boldsymbol{w}}^T P^T \boldsymbol{w} &= \overline{\boldsymbol{w}}^T A \boldsymbol{w} \\
\left(\frac{1}{1-\lambda} + \frac{1}{1-\overline{\lambda}} - 1\right)\overline{\boldsymbol{w}}^T A \boldsymbol{w} &= \overline{\boldsymbol{w}}^T (P + P^T - A)\boldsymbol{w}
\end{aligned}$$

Proof (2): Since

$$\frac{1}{1-\lambda}+\frac{1}{1-\bar{\lambda}}-1=\frac{1-|\lambda|^2}{|1-\lambda|^2}\in\mathbb{R}$$

is real, but $\boldsymbol{w}$ may be complex $\boldsymbol{w}=\boldsymbol{w}_{\mathrm{R}}+i\boldsymbol{w}_{\mathrm{I}}$, we take the real part

$$\mathrm{Re}\left\{\left(\frac{1}{1-\lambda}+\frac{1}{1-\bar{\lambda}}-1\right)\overline{\boldsymbol{w}}^T A\boldsymbol{w}\right\}=\mathrm{Re}\left\{\overline{\boldsymbol{w}}^T(P+P^T-A)\boldsymbol{w}\right\}$$

We compute

$$\begin{aligned}\mathrm{Re}\left\{\overline{\boldsymbol{w}}^T A\boldsymbol{w}\right\}&=\mathrm{Re}\left\{(\boldsymbol{w}_{\mathrm{R}}-i\boldsymbol{w}_{\mathrm{I}})^T A(\boldsymbol{w}_{\mathrm{R}}+i\boldsymbol{w}_{\mathrm{I}})\right\}=\boldsymbol{w}_{\mathrm{R}}^T A\boldsymbol{w}_{\mathrm{R}}+\boldsymbol{w}_{\mathrm{I}}^T A\boldsymbol{w}_{\mathrm{I}}\\ \mathrm{Re}\left\{\overline{\boldsymbol{w}}^T(P+P^T-A)\boldsymbol{w}\right\}&=\boldsymbol{w}_{\mathrm{R}}^T(P+P^T-A)\boldsymbol{w}_{\mathrm{R}}+\boldsymbol{w}_{\mathrm{I}}^T(P+P^T-A)\boldsymbol{w}_{\mathrm{I}}\end{aligned}$$

So that

$$\frac{1-|\lambda|^2}{|1-\lambda|^2}\left(\boldsymbol{w}_{\mathrm{R}}^T A\boldsymbol{w}_{\mathrm{R}}+\boldsymbol{w}_{\mathrm{I}}^T A\boldsymbol{w}_{\mathrm{I}}\right)=\boldsymbol{w}_{\mathrm{R}}^T(P+P^T-A)\boldsymbol{w}_{\mathrm{R}}+\boldsymbol{w}_{\mathrm{I}}^T(P+P^T-A)\boldsymbol{w}_{\mathrm{I}}$$

Proof (3): Both A and $P + P^T - A$ are positive definite and it is impossible for both $\boldsymbol{w}_{\mathrm{R}}$ and $\boldsymbol{w}_{\mathrm{I}}$ to vanish simultaneously, since this would imply $\boldsymbol{w} = \mathbf{0}$. Hence,

$$\boldsymbol{w}_{\mathrm{R}}^T A \boldsymbol{w}_{\mathrm{R}} + \boldsymbol{w}_{\mathrm{I}}^T A \boldsymbol{w}_{\mathrm{I}} > 0$$
$$\boldsymbol{w}_{\mathrm{R}}^T (P + P^T - A) \boldsymbol{w}_{\mathrm{R}} + \boldsymbol{w}_{\mathrm{I}}^T (P + P^T - A) \boldsymbol{w}_{\mathrm{I}} > 0$$

Therefore,

$$\frac{1 - |\lambda|^2}{|1 - \lambda|^2} > 0$$

Hence, $|\lambda| < 1$ for any $\lambda \in \sigma(H)$. Consequently, $\rho(H) < 1$ and the iterative scheme (3) converges. $\square$

Jacobi, Gauss-Seidel, and SOR methods

Any matrix A can be split as $A = D - L_0 - U_0$, where D is diagonal, L_0 is strictly lower-triangular, and U_0 is strictly upper-triangular. Let $L = D^{-1}L_0$ and $U = D^{-1}U_0$. Then, an iterative scheme $\boldsymbol{x}_{k+1} = H\boldsymbol{x}_k + \boldsymbol{v}$ can be constructed in several new ways:

- *Jacobi* method: $H = L + U$ and $\boldsymbol{v} = D^{-1}\boldsymbol{b}$ or as regular splitting

$$D\boldsymbol{x}_{k+1} = (L_0 + U_0)\boldsymbol{x}_k + \boldsymbol{b}$$

- *Gauss-Seidel* method: $H = (I - L)^{-1}U$ and $\boldsymbol{v} = (I - L)^{-1}D^{-1}\boldsymbol{b}$ or as regular splitting

$$(D - L_0)\boldsymbol{x}_{k+1} = U_0\boldsymbol{x}_k + \boldsymbol{b}$$

- *Successive Over-Relaxation* (SOR) method: $H_\omega = (I - \omega L)^{-1}[(1 - \omega)I + \omega U]$ and $\boldsymbol{v}_\omega = \omega(I - \omega L)^{-1}D^{-1}\boldsymbol{b}$, where $\omega \in [1, 2)$ is the *relaxation parameter*. As regular splitting:

$$\left(\frac{1}{\omega}D - L_0\right)\boldsymbol{x}_{k+1} = \left[\left(\frac{1}{\omega} - 1\right)D + U_0\right]\boldsymbol{x}_k + \boldsymbol{b}$$

Lemma (Gerschgorin): *Let $A \in \mathbb{C}^{n\times n}$ be an an arbitrary matrix with elements $a_{k,m}$. Then the following holds for the spectrum of A:*

$$\sigma(A) \subset \cup_{k=1}^{n} \mathbb{S}_k \tag{4}$$

$$\mathbb{S}_k = \left\{ z \in \mathbb{C} \ : \ |z - a_{k,k}| \leq \sum_{m \neq k}^{n} |a_{k,m}| \right\} \tag{5}$$

Moreover, $\lambda \in \sigma(A)$ may lie on the boundary $\partial\mathbb{S}_\ell$ for some $\ell \in \{1, 2, \ldots, n\}$ only if $\lambda \in \partial\mathbb{S}_k$ for all $k \in \{1, 2, \ldots, n\}$. The $\mathbb{S}_k$ are known as Gerschgorin disks.

Definition: A matrix A is (strictly) *diagonally dominant* if

$$|a_{k,k}| \geq \sum_{m \neq k}^{n} |a_{k,m}|, \quad k = 1, 2, \ldots, n \tag{6}$$

with strict inequality for at least one row.

Theorem: *If the matrix A is diagonally dominant, then both the Jacobi and Gauss-Seidel methods converge.*

Proof (1): *Jacobi:* The iteration matrix of the Jacobi method is $H = D^{-1}L_0 + D^{-1}U_0$, where $d_{k,k} = a_{k,k}$. Hence,

$$\sum_{m=1}^{n} |h_{k,m}| = \sum_{m \neq k}^{n} |h_{k,m}| = \frac{1}{|a_{k,k}|} \sum_{m \neq k}^{n} |a_{k,m}| \leq 1, \quad k = 1, 2, \ldots, n$$

and this inequality is sharp for at least one k. Since $h_{k,k} = 0$, by the Gerschgorin Lemma, $\rho(H) < 1$ and the Jacobi iteration converges.

Gauss-Seidel: For the Gauss-Seidel method: $H = (I - D^{-1}L_0)^{-1}D^{-1}U_0$, $\boldsymbol{v} = (I - D^{-1}L_0)^{-1}D^{-1}\boldsymbol{b}$. Choose $\lambda \in \sigma(H)$ with the corresponding eigenvector $\boldsymbol{w}$. Hence,

$$\begin{aligned} H\boldsymbol{w} &= \lambda \boldsymbol{w} \\ (I - D^{-1}L_0)^{-1}D^{-1}U_0\boldsymbol{w} &= \lambda \boldsymbol{w} \\ U_0\boldsymbol{w} &= \lambda D(I - D^{-1}L_0)\boldsymbol{w} \\ U_0\boldsymbol{w} &= \lambda (D - L_0)\boldsymbol{w} \end{aligned}$$

Proof (2): Since $A = D - L_0 - U_0$, we can rewrite $U_0\boldsymbol{w} = \lambda(D - L_0)\boldsymbol{w}$ as

$$\begin{aligned}
-\sum_{m=k+1}^{n} a_{k,m} w_m &= \lambda a_{k,k} w_k + \lambda \sum_{m=1}^{k-1} a_{k,m} w_m \\
\lambda a_{k,k} w_k &= -\sum_{m=k+1}^{n} a_{k,m} w_m - \lambda \sum_{m=1}^{k-1} a_{k,m} w_m \\
|\lambda|\,|a_{k,k}|\,|w_k| &\leq \sum_{m=k+1}^{n} |a_{k,m}|\,|w_m| + |\lambda| \sum_{m=1}^{k-1} |a_{k,m}|\,|w_m| \\
&\leq \left(\sum_{m=k+1}^{n} |a_{k,m}| + |\lambda| \sum_{m=1}^{k-1} |a_{k,m}| \right) \max_m |w_m|
\end{aligned}$$

for all $k = 1, 2, \ldots, n$

Proof (3): Let q be the index for which $|w_q| = \max_m |w_m| > 0$. Then,

$$|\lambda|\,|a_{q,q}| \leq \sum_{m=q+1}^{n} |a_{q,m}| + |\lambda| \sum_{m=1}^{q-1} |a_{q,m}|$$

Assume that $|\lambda| \geq 1$. In which case

$$|a_{q,q}| \leq \frac{1}{|\lambda|} \sum_{m=q+1}^{n} |a_{q,m}| + \sum_{m=1}^{q-1} |a_{q,m}| \leq \sum_{m=q+1}^{n} |a_{q,m}| + \sum_{m=1}^{q-1} |a_{q,m}| \leq \sum_{m \neq q}^{n} |a_{q,m}|$$

However, since also $|a_{q,q}| \geq \sum_{m \neq q}^{n} |a_{q,m}|$, it must be that $|a_{q,q}| = \sum_{m \neq q}^{n} |a_{q,m}|$. At the same time, as we showed earlier

$$|\lambda|\,|a_{k,k}|\,|w_k| \leq \sum_{m=k+1}^{n} |a_{k,m}|\,|w_m| + |\lambda| \sum_{m=1}^{k-1} |a_{k,m}|\,|w_m|, \quad k = 1, 2, \ldots, n$$

Proof (4): Considering $k = q$ and substituting $|a_{q,q}| = \sum_{m \neq q}^{n} |a_{q,m}|$ we get

$$|\lambda| \, |a_{q,q}| \, |w_q| \leq \sum_{m=q+1}^{n} |a_{q,m}| \, |w_m| + |\lambda| \sum_{m=1}^{q-1} |a_{q,m}| \, |w_m|$$

$$|\lambda| \sum_{m \neq q}^{n} |a_{q,m}| \, |w_q| \leq \sum_{m=q+1}^{n} |a_{q,m}| \, |w_m| + |\lambda| \sum_{m=1}^{q-1} |a_{q,m}| \, |w_m| \leq |\lambda| \sum_{m \neq q}^{n} |a_{q,m}| \, |w_m|$$

$$\sum_{m \neq q}^{n} |a_{q,m}| \, |w_q| \leq \sum_{m \neq q}^{n} |a_{q,m}| \, |w_m|$$

Since $|w_q| = \max_m |w_m|$, the last inequality can only hold if $|w_m| = |w_q|$ for all $m = 1, 2, \ldots, n$. Hence, any $k = 1, 2, \ldots, n$ can be chosen as q and the equality $|a_{k,k}| = \sum_{m \neq k}^{n} |a_{k,m}|$ must hold for all $k = 1, 2, \ldots, n$, which contradicts the assumption of strict diagonal dominance. Therefore, $|\lambda| < 1$ and $\rho(H) < 1$. $\square$

Classical iterative methods: conclusion

- To solve $A\boldsymbol{x} = \boldsymbol{b}$
- One could use various regular splittings $A = P - N$
- And iterations $P\boldsymbol{x}_{k+1} = N\boldsymbol{x}_k + \boldsymbol{b}$, which is the same as $\boldsymbol{x}_{k+1} = P^{-1}N\boldsymbol{x}_k + P^{-1}\boldsymbol{b}$
- There are several splittings, which guarantee $\rho(P^{-1}N) < 1$, i.e. convergence
- Sometimes a symmetry of A is assumed, convergence may be slow